

SEC P vs NP log 2026-05-17

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 00:27 UTC

SEC P vs NP — daily log 2026-05-17

2 cycles recorded

GCT Lie-Stabilizer Codimension Linearly Bounds ACC^0 Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric Complexity Theory: Lie-algebra stabilizers of multilinear polynomial extensions in the Mulmuley-Sohoni-Bürgisser-Ikenmeyer framework $\times \text{ACC}^0[m]$ circuit size for Boolean functions (Sipser-like targets)
- **Recorded:** 2026-05-17 00:01 UTC
- **Entry ID:** 83d26c2ef069

Statement

For a Boolean function $f: \{-1, +1\}^n \rightarrow \{-1, +1\}$, let $F_f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \cdot \prod_{i \in S} x_i$ be its multilinear Walsh extension over $\mathbb{Q}$, let $L(F_f) := \{A \in \mathfrak{gl}_n(\mathbb{Q}) : \sum_{i,j} A_{ij} x_j \partial_i F_f \equiv 0 \text{ in } \mathbb{Q}[x_1, \dots, x_n]\}$ be its GCT Lie stabilizer, and set $\gamma(f) := n^2 - \dim_{\mathbb{Q}} L(F_f)$. Conjecture: there is an absolute constant $C \leq 10$ such that every $\text{ACC}^0[m]$ circuit of size s computing f satisfies $\gamma(f) \geq n^2 - C \cdot s$. A single $\text{ACC}^0[m]$ circuit of size s computing some f with $\gamma(f) < n^2 - 10 \cdot s$ falsifies the conjecture.

Rationale

GCT's symmetry-characterization principle says polynomials whose stabilizer Lie algebras exceed generic dimension are atypical; we invert this to claim small ACC^0 circuits cannot accidentally synthesize multilinear extensions of anomalously high symmetry. The Lie stabilizer is GL_n -equivariant rather than truth-table-dense, so it dodges Razborov-Rudich's 'large + constructive' clause, and it is computed from the GL_n action on polynomials, not from black-box algebraic extensions — placing the invariant outside the algebrization and relativization regimes that GCT was designed to escape.

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 102, in <module>
    result = run_trial(seed)
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 86, in run_trial
    gamma = gaussian_elimination(system)
    ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 65, in gaussian_elim
    if matrix[i][i] == 0:
    ~~~~~^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` in `gaussian_elimination` before any of the 480 trials or reference-function checks produced slack data, so the pre-registered support condition cannot be evaluated. Per Rule 2, a crash yields **INCONCLUSIVE**. | next: Fix the Gaussian elimination routine (guard against out-of-range pivot indexing and handle rectangular/short matrices) and rerun the full (n,s)-grid plus reference-function suite to obtain actual slack statistics.

2-adic Newton Slopes of DISJ Sub-blocks Scale as $\sqrt{n}$

- **Verdict:** INCONCLUSIVE
- **Bridge:** p-adic analysis (Newton polygons over $\mathbb{Q}_2$ of integer characteristic polynomials) $\times$ Randomized communication complexity of DISJOINTNESS via average-case tensor invariants on sub-blocks
- **Recorded:** 2026-05-17 00:26 UTC
- **Entry ID:** efd166108501

Statement

Let $M_n \in \{-1, +1\}^{\{2^n \times 2^n\}}$ be the signed disjointness matrix $M_n[x, y] = (-1)^{|x \cap y|}$ where x, y are interpreted as subsets of $[n]$. For each n , draw uniformly random row set R and column set C with $|R|=|C|=k=\min(2n, 40)$, form $A := M_n[R, C]$, compute its exact integer characteristic polynomial $p(t) = \sum_{i=0}^k c_i t^i$, and define the 2-adic Newton slope count $v_2(A) :=$ number of distinct slopes in the lower convex hull of $\{(i, v_2(c_i)) : c_i \neq 0\}$ where v_2 is the 2-adic valuation ($v_2(0) := +\infty$). Conjecture: $E_{R, C}[v_2(M_n^{\{R, C\}})] = \Theta(\sqrt{n})$; moreover for every $n \geq 8$ the DISJ median exceeds the median of v_2 for an i.i.d. ± 1 matrix of the same shape by a factor ≥ 1.5 , and a single (n, seed) pair where this median ratio inverts falsifies the conjecture.

Rationale

The subset semilattice underlying DISJ injects 2-adic congruence structure into every sub-block (parity of intersection size acts like a $\mathbb{Z}_2$ -valued character), so subdeterminants accumulate distinguished 2-adic valuations that random Bernoulli matrices lack — a property that should propagate, via Yao’s principle, into a lower bound on randomized communication. Newton polygons over $\mathbb{Q}_2$ are not preserved by polynomial extension to $\mathbb{F}_2[t]/(t^2-t)$, evading the algebrization barrier, and the invariant is structural rather than ‘large+useful on truth tables’, evading Razborov-Rudich. The bridge between p-adic Newton-polygon geometry and communication complexity has essentially no prior literature, while the average-to-worst-case framing ($E[v_2] \mapsto$ worst-case CC lower bound) follows the Impagliazzo-Wigderson template.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 34.76s

```
tances_tested': 5, 'conjecture_holds': False, 'counterexample': 'Ratio 1.0 < 1.5 for seed 547'}
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
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TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
```

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Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <module>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                    ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <genexpr>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                    ~~~~~^
```

KeyError: 'seed'

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple trials report conjecture_holds=False with ratio $1.0 < 1.5$ (e.g., seeds 547, 593, 631, ...), satisfying the pre-registered falsification condition | next: Re-run with corrected seed-logging, stratify by $n \in \{10, 14, 18, 22\}$, and check whether the ν_2 statistic saturates at small $k = \min(2n, 40)$ — perhaps a normalized slope-density (ν_2/k) would better capture any DISJ-vs-random gap.